AI WRITING DETECTION REPORT

Sentence-Level Statistical & Linguistic Analysis

Submission ID: trn:oid:::1:2023323351
Report Date: 10/24/2026, 10:26:03 PM

Submission Title	CHAPTER 1	**Word Count**	9541
Author	Student Submission	**Sentence Burstiness (CV)**	0.51 (Natural Variance)
Lexical Diversity (MATTR)	86.6%	**Shannon Entropy**	9.13 bits

How much of this submission has been generated by AI?

10% of qualifying prose text in this submission has been determined to be likely generated by AI or modified using an AI paraphraser. (Scores between 1% and 19% are displayed as ***%** to prevent low-threshold false positives.)

***%**

DETECTED AS AI

3%

AI-generated only (Cyan Highlights)

Text exhibiting uniform sentence burstiness, predictable lexical entropy, and direct LLM structural patterns.

7%

AI-generated & AI-paraphrased (Purple Highlights)

Text exhibiting high synonym-substitution density while retaining uniform AI clause skeletons.

SUBMISSION BREAKDOWN BAR

SUBMITTED DOCUMENT TEXT (CYAN = AI-GENERATED • PURPLE = AI-PARAPHRASED)

CHAPTER 1 INTRODUCTION A STUDY OF POTENTIAL OF GREEN FINANCE IN INDIA WITH SPECIAL REFERENCE TO UTTARAKHAND 1. 1 Background of the Study Economic development depends on the allocation of capital across competing activities, regions and periods of time. That allocation becomes a matter of public concern when profitable investments impose environmental costs on others or when socially beneficial investments remain unfunded because their returns cannot be captured in ordinary financial transactions. Climate change brings both difficulties into sharp relief. Rising

temperatures, changing rainfall, ecological degradation and more severe hazards threaten production systems, infrastructure and household security, while the investments needed to reduce emissions and strengthen resilience often demand substantial initial expenditure and extended repayment periods (Intergovernmental Panel on Climate Change, 2023). The question is not simply whether finance exists. It is whether finance reaches suitable activities on conditions that reflect their wider economic and environmental value. The economic case for timely action rests partly on the possibility that the costs of delayed environmental response may exceed the costs of well designed preventive investment. Stern (2007) treats climate change as a persistent failure of conventional markets because the damages associated with greenhouse gas emissions extend across borders, generations and sectors. This framing does not imply that every environmentally labelled investment is desirable. It indicates that private prices, when taken alone, may provide an incomplete guide to social welfare. The resulting policy concern is how public institutions, financial intermediaries and investors can redirect resources without overlooking technological uncertainty, affordability or the distribution of costs. Sustainable development places this financing problem within a broader conception of human welfare. The United Nations (2015) links environmental protection with poverty reduction, energy access, resilient infrastructure, sustainable settlements and responsible resource use. These goals do not always move together. A large infrastructure project may supply cleaner electricity while placing pressure on land, rivers or local livelihoods. An environmentally sound investment may remain inaccessible to poorer households if lending conditions require formal collateral or high personal contributions. Green finance therefore cannot be assessed solely through the volume of capital announced. Its purpose, allocation, implementation and social effects also require attention. An important distinction concerns mitigation and adaptation. Mitigation finance supports activities that reduce emissions or increase carbon absorption, while adaptation finance strengthens the ability of people, infrastructure and ecosystems to cope with climatic stress. Available evidence indicates that adaptation continues to attract less finance than the scale of documented need would warrant, particularly in developing economies where protective investments may generate substantial public benefits but limited direct cash returns (United Nations Environment Programme, 2024). This imbalance has immediate relevance for mountain regions, where the financial consequences of extreme rainfall, slope instability, water stress and disrupted connectivity can be considerable even when local emissions remain comparatively modest. India illustrates the difficulty of aligning national development requirements with environmental commitments. Expanding electricity access, supporting productive employment, improving transport and strengthening agricultural incomes require

employment, improving transport and strengthening agricultural incomes require sustained investment. At the same time, India's updated nationally determined contribution identifies lower emissions intensity, greater nonfossil electricity capacity and stronger adaptation in vulnerable sectors, including the Himalayan region, as public priorities (Government of India, 2022). These commitments create demand for financial arrangements capable of supporting renewable energy, clean mobility, resource efficient enterprises, ecosystem protection and agricultural resilience. The effectiveness of such arrangements depends on their operation within particular states rather than on national commitments alone. Uttarakhand provides a particularly instructive setting for this inquiry. Its mountain ecology supports river systems, forests, agriculture, tourism and hydropower, yet the same physical setting exposes settlements and productive assets to climatic and geological risks. Research on the Hindu Kush Himalaya suggests that mountain economies experience environmental change through interconnected pressures on water, ecosystems, mobility and livelihoods, making it difficult to separate ecological security from regional development (Wester et al. , 2019). The presence of valuable natural resources does not automatically produce environmentally responsible investment. Financing decisions must account for access, local conditions, project scale, maintenance needs and the possibility of harmful cumulative effects. Regional differences within Uttarakhand add another layer of complexity. Economic activity, urban services and access to formal financial institutions are not distributed uniformly across plains, urban centres and remote hill districts. Official state publications provide a basis for examining these variations through district statistics, economic indicators and information on productive sectors (Directorate of Economics and Statistics, Government of Uttarakhand, 2025b). From the standpoint of green finance, the relevant question is whether available instruments and institutional arrangements respond to these differences or favour projects that are larger, more visible and easier to administer. Against this background, the present study examines the potential of green finance in India with special reference to Uttarakhand. India supplies the policy and institutional context, while Uttarakhand constitutes the principal unit of analysis. The study gives particular attention to clean energy because it offers an established field in which financial commitments, project development and observable outputs can be examined. It also considers renewable energy, sustainable tourism, clean transportation and climate resilient agriculture as possible areas for wider growth. Throughout the enquiry, the analysis remains exploratory and qualitative, drawing on existing documentary evidence rather than presuming outcomes that the available records have not established. 1. 2 Concept and Scope of Green Finance Green finance refers to financial resources and arrangements directed toward activities that generate identifiable environmental benefits

These arrangements include public expenditure, grants, bank loans, equity investment, credit guarantees, concessional finance, dedicated deposits, bonds and other forms of financial intermediation.

that a useful definition must consider both the financing of environmental goods and the financing of policies that encourage their provision. The concept therefore concerns more than a particular financial product. It refers to the relationship between the purpose of finance and the environmental character of the activity being financed. An operational definition becomes necessary because common terms in this field overlap without being interchangeable. Climate finance concentrates on mitigation and adaptation. Green finance covers climate related activities but may also include pollution prevention, biodiversity protection, waste management and other environmental objectives. Sustainable finance has a wider reach because it considers environmental concerns together with social and governance questions. Investment guided by environmental, social and governance factors may assess financial risk and corporate conduct without necessarily directing new capital to a demonstrably green project. Green investment refers to the physical or productive expenditure itself, while green finance refers to the resources and instruments that make that expenditure possible (Eyraud et al. , 2013). The Reserve Bank of India provides a particularly relevant institutional definition for the Indian setting. Its framework for green deposits describes green finance as lending or investment that contributes to climate mitigation, adaptation, resilience or other environmental objectives, including biodiversity management and nature related solutions (Reserve Bank of India, 2023a). This definition is useful because it connects the environmental purpose of finance with identifiable activities and reporting requirements. It also indicates that a financial product cannot be treated as green solely because its marketing materials use environmental language. Table 1. 1 Conceptual distinctions relevant to the study Concept Principal focus Implication for this thesis Green finance Finance directed toward documented environmental objectives. Includes eligible public expenditure, credit, investment and related instruments. Climate finance Mitigation of climate change and adaptation to its effects. Forms one important component of the broader green finance enquiry. Sustainable finance Environmental, social and governance considerations together. Included only where an identifiable environmental financial purpose is present. Green investment Physical or productive expenditure on environmentally beneficial activity. Represents the financed activity rather than the financing instrument itself. ESG finance Investment or lending informed by environmental, social and governance criteria. Not assumed to be green without evidence concerning allocation and use of funds. Differences in classification matter for empirical

research. A bank loan for an ordinary commercial building does not become green merely because the borrower also undertakes unrelated environmental activities. A solar installation financed from a conventional loan may qualify as green in substance even when the lender does not market the product as a green loan, provided the environmental purpose and use of proceeds can be established. Public spending on watershed restoration may count as environmentally relevant finance even where it produces no conventional financial return. Conversely, a labelled bond may have limited additional environmental value if its proceeds merely refinance expenditure that would have occurred in any event (Flammer, 2021). The scope adopted in this study is therefore based on purpose, evidence and institutional context. Eligible activities include clean and renewable energy, energy efficiency, environmentally responsible tourism, clean

Supporting investments in storage, electricity networks, charging facilities, water conservation, waste management and ecological protection are relevant when their connection to these sectors can be documented.

reporting practices and risk management arrangements that influence whether such activities receive finance (Wang & Zhi, 2016). Three cautions follow. First, the environmental suitability of a project must be assessed in relation to its local effects, not inferred from its sector alone. Second, an announced allocation should not be equated with funds actually disbursed. Third, financial deployment should not be confused with verified environmental improvement. These distinctions are particularly important in Uttarakhand, where renewable energy and tourism may offer economic benefits while also generating ecological pressure if project design, location or scale are inappropriate. 1. 3 Economic Rationale for Green Finance The economic rationale for green finance begins with externalities. Environmental damage often imposes costs on people who are not parties to the transaction that produces it. Clean air, stable climatic conditions, functioning watersheds and ecosystem protection also display characteristics associated with public goods, meaning that private investors may struggle to recover the full social benefits of protecting them. Coase (1960) demonstrates that the treatment of social costs depends on property rights, transaction costs and institutional arrangements. In practice, environmental negotiation across dispersed communities and future generations is rarely

Public policy and specialised financial arrangements may therefore be required to influence the direction of investment.

ot settle the question of which intervention is appropriate. Gillingham and Stock (2018) show that the costs of reducing emissions vary across technologies and policy choices. Some projects may deliver environmental benefits at comparatively low cost, while others

A sound approach to green finance must therefore distinguish genuine social value from projects that obtain favourable treatment without adequate environmental justification. an intervention suitable for a national utility may not be suitable for a small agricultural producer or a remote tourism enterprise. Information problems provide a second rationale. Lenders may find it difficult to judge the technical quality, projected cash flow, environmental performance or repayment capacity of unfamiliar green projects. Akerlof (1970) explains how uneven information about quality can weaken markets when buyers cannot reliably distinguish strong propositions from weak ones. In green finance, similar difficulties arise when a bank cannot establish whether a proposed solar system will function as expected, whether savings will materialise, or whether an environmental label is credible. The result may be higher perceived risk, additional documentation requirements or the rejection of otherwise useful projects. Credit rationing can persist even where borrowers are willing to pay a higher interest rate. Stiglitz and Weiss (1981) argue that, under imperfect information, raising the interest rate can change the pool of borrowers and the incentives surrounding repayment, giving lenders reasons to restrict credit rather than clear the market through price. This insight is relevant for households, farmers and small enterprises in Uttarakhand that may possess limited collateral, Their inability to obtain green credit should not automatically be interpreted as evidence that the underlying investment lacks Renewable energy adds a distinctive financial consideration because many projects combine substantial initial capital expenditure with comparatively low operating costs. many years, the cost and availability of finance become central determinants of viability. Egli et al. (2018) show that financing conditions are not fixed and may improve as The implication is that the commercial attractiveness of a solar installation cannot be assessed Interest rates, loan tenor, payment security, insurance and operating conditions can materially alter the result. possible response, particularly for larger renewable energy investments. Such arrangements place emphasis on the expected cash flows of a project rather than relying exclusively on the balance sheet of its sponsor. Steffen (2018) finds that project finance has considerable importance in renewable energy development, although its application depends on project size, contractual arrangements and the distribution of risk. For smaller or dispersed projects, transaction costs may make this structure unsuitable unless similar investments can be aggregated. A village energy system and a large grid connected

plant should not be expected to require identical financing terms. Public institutions can address several of these barriers, but their contribution extends beyond direct subsidy. State investment banks may provide patient capital, coordinate technical assessment, improve investor confidence, support project preparation or absorb particular risks that private lenders hesitate to carry (Geddes et al. , 2018). Mazzucato and Semieniuk (2018) likewise suggest that the identity of the investor and the structure of finance influence the kinds of renewable energy projects that proceed. The important question is not whether public or private finance is inherently superior. It is whether their respective roles fit the uncertainties, developmental priorities and social benefits associated with the investment. Policy credibility also affects the willingness of private investors to commit resources. Polzin et al. (2019) argue that renewable energy policies influence investment through revenue expectations, regulatory stability, transaction costs and risk allocation. A favourable policy announcement can attract initial interest, but inconsistent implementation, delayed approvals or uncertainty over tariffs may weaken its practical effect. This distinction is especially relevant for state level research because national financial rules do not automatically resolve local issues relating to land, grid connection, departmental coordination or project maintenance. Green bonds illustrate both the promise and the limitations of financial innovation. Studies of corporate issuance suggest that credible disclosure and external verification can influence whether bond proceeds are associated with meaningful environmental action (Flammer, 2021). Research on investor preferences also indicates that some investors value the environmental characteristics of bonds, though any pricing benefit depends on market conditions and

For developing economies and smaller public authorities, the costs of certification, monitoring and minimum issuance size may restrict access to this instrument (Banga, 2019).

market therefore does not guarantee that small projects in Uttarakhand can draw on it directly. Adaptation introduces further complications because financial returns may appear as losses avoided rather than revenue earned. A protected water source, a stabilised slope or a resilient agricultural system can generate substantial social benefits without producing a dedicated stream of cash for debt servicing. The gap identified by the United Nations Environment Programme (2024) suggests that conventional commercial criteria may leave adaptation needs underfunded even where the economic case for

Grants, budgetary allocations, concessional credit, insurance and public development finance may consequently assume greater importance in these activities.

Financial resources may flow toward firms with formal records, established assets and

proximity to urban markets, while remote communities with high environmental vulnerability remain underserved. This pattern can occur without explicit discrimination because standard lending procedures reward lower transaction costs and predictable

Yet an allocation that appears efficient from the perspective of an individual lender may be inadequate from the perspective of balanced regional development. Assessment of green finance in Uttarakhand should therefore consider who receives support, in which districts, under what terms, and with what documented results. ken together, these arguments suggest that green finance is not merely a specialised branch of investment management. It represents an institutional response to environmental externalities, incomplete information, uncertain returns, long investment horizons and uneven access to capital. Whether that response is effective remains an empirical question. The answer depends on the quality of projects, the credibility of policy, the suitability of financial instruments and the ability of public and private institutions to translate commitments into observable activity. 1. 4 Green Finance in India India's engagement with green finance arises from the interaction between development priorities and international climate commitments. Its updated nationally determined contribution commits the country to reduce the emissions intensity of gross domestic product by 45 per cent by 2030 relative to the 2005 level. It also sets an objective of achieving about 50 per cent of cumulative electric power installed capacity from nonfossil energy resources by 2030, subject to support that includes technology transfer and affordable international finance (Government of India, 2022). The installed capacity commitment should not be confused with a claim that 50 per cent of electricity generation or total energy consumption must come from those sources. The official document also explicitly identifies agriculture, water resources, the Himalayan region, health and disaster management as areas requiring stronger adaptation investment. These commitments create financing requirements across electricity generation, transmission, storage, transport, agriculture, urban infrastructure and ecological protection. The demand cannot be met through a single instrument. Public budgets can support activities with substantial social returns, while banks, development finance institutions and capital markets can finance projects with identifiable repayment prospects. Grants, concessional loans, credit guarantees, viability support and blended arrangements may also be relevant where risks or social benefits prevent conventional commercial lending from operating effectively (Reserve Bank of India, 2023b). The composition of finance matters because the needs of a large solar park differ from those of a small rooftop installation or a watershed protection programme. India's sovereign green bond framework, introduced in 2022, provides an institutional basis for linking government borrowing with eligible public expenditure on environmentally beneficial

government borrowing with eligible public expenditure on environmentally beneficial projects. The framework specifies processes for project evaluation, management of proceeds and reporting, thereby offering a means of identifying the public purposes associated with the instrument (Department of Economic Affairs, Government of India, 2022). Such borrowing may broaden attention to environmental expenditure, but the size of an issuance does not establish that every supported project is additional, effective or

State level research must examine the actual destination and use of public resources before attributing local outcomes to a national borrowing programme.

en deposits in April 2023, with effect from June 2023. The framework applies to specified regulated entities that offer such deposits and sets requirements concerning eligible activities, allocation, verification and reporting (Reserve Bank of India, 2023a). It should not be interpreted as a direction compelling every bank to introduce a green deposit product. For Uttarakhand, the relevant question is whether participating institutions translate these national provisions into accessible credit for renewable energy, clean transport, resilient agriculture or other qualifying activities within the state. Capital market regulation provides a further institutional channel. The Securities and Exchange Board of India revised disclosure requirements for the issuance and listing of green debt securities in 2023, placing attention on the use of proceeds, project evaluation and continuing disclosure (Securities and Exchange Board of India, 2023). These requirements may improve the ability of investors to examine environmental claims, although compliance does not by itself prove environmental success. In June 2025, the regulator introduced a separate framework for environmental, social and governance debt securities other than green debt securities (Securities and Exchange Board of India, 2025). Because this measure falls after the proposed core study period ending in the financial year 2024–25, it is treated only as a later contextual development rather than as evidence within the principal historical series. The institutional system extends beyond commercial banks and securities markets. The Indian Renewable Energy Development Agency supports renewable energy finance, while the National Bank for Agriculture and Rural Development works through rural credit, development programmes and related institutional arrangements. NABARD's annual reporting provides a documentary source for examining the financial and developmental context of rural investment, although a national report should not be assumed to reveal complete district level allocations for Uttarakhand (National Bank for Agriculture and Rural Development, 2025). State agencies, electricity utilities, regulatory bodies, cooperative institutions and departmental programmes also influence whether national instruments reach projects on the ground. India's renewable energy statistics offer an important source for distinguishing installed

India's renewable energy statistics offer an important source for distinguishing installed capacity from electricity actually generated. The Ministry of New and Renewable Energy describes its 2024–25 publication as a secondary statistical compilation covering installed capacity and generation at national and state levels, with historical financial years extending back to 2014–15 (Ministry of New and Renewable Energy, Government of India, 2025). This distinction is essential for evaluating clean energy. Installed capacity measures the rated ability to produce power, while generation reflects actual output during a period. Neither measure, on its own, establishes the financial cost of a project, the amount of credit disbursed, the quality of service or the distribution of local benefits. Despite institutional progress, several limitations remain relevant to the study. Financial products may be concentrated in larger projects and established corporate borrowers. Environmental categories may differ across departments or lenders, making aggregation difficult. State level and district level data may not consistently distinguish sanction from disbursement, public contribution from private borrowing, or project approval from operation. Small enterprises and rural households may face collateral requirements, limited technical support and uncertainty about scheme eligibility. These are not conclusions about the present condition of Uttarakhand. They identify questions that require examination against available records. The Indian context therefore establishes the framework within which the state enquiry proceeds. National commitments and financial regulations explain why green investment has become a matter of economic policy, but they cannot answer whether a particular Himalayan state receives suitable finance or achieves meaningful results. That assessment requires attention to state institutions, local sectors, district variation and the relationship between financial support and project performance. 1. 5 Uttarakhand Context Uttarakhand is a Himalayan state whose economic opportunities and environmental pressures are closely connected. Forests, river systems, agricultural land, pilgrimage routes and natural settings contribute to livelihoods and public revenue, while steep terrain, fragile slopes and climatic variability complicate infrastructure development. The study area cannot be understood adequately through statewide averages alone. Official economic and statistical publications provide material for examining differences in population, productive activity, infrastructure and district conditions, although the availability and comparability of individual indicators require careful checking (Directorate of Economics and Statistics, Government of Uttarakhand, 2025a). The ecological importance of mountain regions extends beyond their administrative boundaries. Research on major mountain water systems indicates that their contribution to downstream water security is accompanied by substantial vulnerability to climatic and socioeconomic pressures (Immerzeel et al. , 2020). For Uttarakhand, this suggests that investment decisions

affecting glaciers, catchments, rivers, forests and settlements may have consequences that standard project accounts do not fully capture. The financial assessment of a development activity should therefore consider environmental suitability and possible downstream effects alongside expected revenue. Disaster exposure reinforces this requirement. The 2021 Chamoli disaster demonstrated how a large rock and ice avalanche could trigger destructive downstream impacts and damage infrastructure in a sensitive mountain setting (Shugar et al. , 2021). That event should not be reduced to a

It illustrates why project location, geological conditions, cumulative environmental effects, insurance, emergency

An investment may appear profitable under ordinary operating assumptions yet remain

Clean energy represents a major area of interest because Uttarakhand possesses resources and policy programmes relevant to solar power, small hydropower and decentralised energy

The state solar policy of 2023 and related solar schemes provide documentary evidence of policy intent and programme design (Uttarakhand Renewable Energy Development Agency, 2023).

hydropower similarly indicates formal interest in smaller electricity projects (Uttarakhand Renewable Energy Development Agency, 2015). Neither policy document establishes the extent of actual financing, project completion or reliable generation. Those matters depend on administrative records, project data and other verifiable

Existing agency records nevertheless indicate that the subject is not purely hypothetical.

commissioned micro hydropower projects with a combined capacity of 4. 29 MW and refers to electricity access extending to more than 300 villages and hamlets. Its public project lists identify locations in districts including Bageshwar, Chamoli, Pithoragarh and Uttarkashi (Uttarakhand Renewable Energy Development Agency, n. d.). These figures indicate the existence of a decentralised project base, but the webpage does not clearly date the aggregate statement or establish current operating status, financial disbursement or annual generation. They should therefore be treated as indicative administrative evidence rather than as verified results for a particular financial year.

Decentralised solar installations may be relevant for households, public buildings, small enterprises and remote settlements where local conditions favour distributed supply.

Small hydropower may provide another option where water availability, ecological safeguards and maintenance arrangements support its suitability. The viability of either technology depends on factors such as resource conditions, system design, connection to

the electricity network, storage, seasonal variation, financing terms and the technical capacity available for operation. A state with renewable resources does not automatically possess a complete or equitable investment pipeline. Electricity system requirements also shape the value of clean energy finance. Grid connection, metering, distribution infrastructure, power purchase arrangements and the financial condition of utilities can

A renewable installation that is financed but not connected provides a different outcome from one that generates electricity reliably and supports local economic activity.

energy financing indicates that contractual structure, financing costs and policy stability can materially affect investment performance (Steffen, 2018). These considerations are especially relevant where dispersed settlements and difficult terrain increase infrastructure costs. Tourism constitutes another important area for enquiry. Pilgrimage, nature based travel, hospitality and associated transport support economic activity in Uttarakhand, but visitor growth can increase pressure on water, waste systems, energy use and local infrastructure. Research on the global carbon footprint of tourism indicates that environmental impacts extend through transport, accommodation, food and supply chains rather than being confined to the immediate destination (Lenzen et al. , 2018). Green finance may therefore be relevant to efficient buildings, renewable energy use, waste treatment, water conservation and responsible local enterprises, provided that environmental claims are supported by evidence. The financing requirements of sustainable tourism differ from those of utility scale energy. Many relevant enterprises may be smaller, seasonal or geographically dispersed, with limited access to formal credit. A hotel installing solar water heating, a community enterprise improving waste management and a local transport operator replacing an older vehicle face different combinations of cost, risk and repayment capacity. The existence of tourism activity alone does not establish green finance potential. The study must consider whether environmentally beneficial investments can be identified, financed and monitored within realistic institutional conditions. Clean transportation presents a related set of opportunities and constraints. Uttarakhand's electric vehicle policy of 2019 indicates that the state had formally considered electric mobility within the period relevant to the study (Government of Uttarakhand, 2019). Broader evidence on electric mobility points to the importance of vehicle affordability, charging access, electricity supply, battery performance and supporting policy for the adoption of cleaner transport systems (International Energy Agency, 2024). Mountain terrain, route length, tourist seasonality

Policy existence should therefore be distinguished from actual vehicle deployment, charging investment

Agriculture in Uttarakhand includes households and communities whose production decisions depend on local water availability, climate, soil conditions and market access.

irrigation, protected cultivation, improved storage, suitable crop practices and risk reduction measures. Lipper et al. (2014) argue that agricultural responses to climate change must account for food security, adaptation and, where possible, emissions reduction, while recognising that suitable interventions differ across locations. In a mountain economy, a technically attractive measure may still fail if credit terms, extension support or maintenance arrangements do not match household circumstances. Rural institutions and development finance become particularly relevant in this context. Programmes associated with agriculture, watershed management, self help groups and rural enterprise may influence whether finance reaches smaller borrowers or supports collective investments. NABARD's reporting provides a national institutional reference point, while state and district records are required to establish the scale and character of activity within Uttarakhand (National Bank for Agriculture and Rural Development, 2025). It would be inappropriate to treat all agricultural lending as green finance without identifying the environmental or resilience purpose of the expenditure. Institutional coordination cuts across all four sectors. Energy agencies, electricity utilities, tourism authorities, transport departments, agricultural institutions, banks and local

The practical challenge is to connect policy intention with sanction, disbursement, implementation and documented outcomes.

merely as an illustration appended to a national discussion. It provides the central setting in which the capacity, reach and constraints of green finance are examined. 1. 6 Origin and Statement of the Research Problem The research problem arises from an apparent separation between the recognised need for environmentally responsible investment and the limited clarity about how such investment is financed within a particular Himalayan state. India has adopted climate commitments, financial regulations and public programmes that refer to environmentally beneficial activities. Uttarakhand possesses sectors in which such activities may be relevant. What remains insufficiently clear is whether the corresponding financial arrangements are identifiable, accessible, appropriately distributed and associated with verifiable outcomes. One part of the problem concerns identification. Green finance may enter Uttarakhand through departmental budgets, central schemes, development finance institutions, commercial bank loans, utility investment, public incentives or private expenditure. These resources are not always recorded under a common environmental category. A solar installation

might appear in an energy programme, a lending statement or a district development report, while a water conservation investment might be recorded only as agricultural or rural expenditure. Without careful documentary interpretation, the absence of a green label can conceal environmentally relevant finance, while the presence of a green label can exaggerate the environmental significance of an ordinary transaction (Ozili, 2022). A second concern relates to the movement from policy to practice. A policy notification indicates formal intention, but it does not establish that funds were sanctioned. A sanction does not establish disbursement. Disbursement does not establish project completion, and completion does not establish continued operation or environmental improvement. These distinctions are central to the study because public discussion frequently treats announced projects, approved capacities or financial targets as equivalent to actual achievement. The problem is therefore not resolved by assembling a list of schemes. It requires analysis of the available evidence at successive stages of implementation. The clean energy sector illustrates this issue clearly. Renewable energy projects may attract public support and investor interest, yet their economic usefulness depends on financing terms, connection to the electricity system, maintenance and actual generation. Official renewable energy statistics distinguish installed capacity from generation, making clear that the two indicators describe different aspects of performance (Ministry of New and Renewable Energy, Government of India, 2025). Where financial information can be located, the study must consider whether observed capacity or generation can reasonably be associated with particular programmes or financing channels without claiming causal certainty that the documentary evidence cannot support. Regional variation introduces a further dimension. Financial institutions may prefer accessible districts, urban markets or larger borrowers because these settings reduce transaction and monitoring costs. Remote hill districts may face different barriers even when their need for decentralised energy, resilient agriculture or environmental protection is considerable. Official district statistics can help identify contextual differences, but they do not automatically reveal district level green finance flows (Directorate of Economics and Statistics, Government of Uttarakhand, 2025b). Establishing whether such variation exists requires careful examination of sectoral and administrative records. The problem also extends beyond clean energy. Sustainable tourism, clean transportation and climate resilient agriculture may present environmentally relevant investment opportunities, yet their financial pathways,

A framework designed around large renewable projects may not address a small hospitality enterprise, a

Sectoral potential must therefore

be assessed in relation to documented needs, existing initiatives, financial instruments and constraints rather than assumed from general environmental desirability.

central research problem may consequently be stated as the limited availability of an integrated and context sensitive account of the potential, operation and observable effects of green finance in Uttarakhand. This problem includes uncertainty about institutional maturity, sectoral and regional variation, the role of clean energy, and the prospects for growth in additional environmentally relevant activities. The study addresses that uncertainty through exploratory examination of existing evidence. It does not assume that finance is either effective or ineffective before the relevant records have been considered. 1. 7 Research Gap Research on green finance has expanded across questions of financial innovation, environmental policy, investment behaviour and institutional

Reviews indicate, however, that the field remains uneven in its definitions, geographic coverage and treatment of implementation (Ozili, 2022).

attention has been given to national commitments, international capital markets and major financial instruments. Less attention has been paid to the practical organisation of green finance within a specific mountain state whose environmental conditions, institutional capacities and regional disparities differ from those assumed in broader national assessments. One gap concerns spatial scale. National studies can identify the growth of renewable energy investment, regulatory frameworks or green bond issuance, but they rarely establish how finance reaches particular districts or communities. Uttarakhand requires a different level of enquiry because access, terrain and ecological

State statistics offer useful contextual information, yet they do not by themselves provide a consolidated account of green financial flows.

limits the ability to judge whether national policy instruments are relevant to local needs. A second gap concerns the relationship between finance and outcomes. Research on renewable energy often examines installed capacity, policy incentives or technology costs, while financial studies may concentrate on instruments, pricing or investor preferences. These approaches answer important but different questions. Work on project finance and financing conditions shows that the structure of capital affects renewable energy investment, but the existence of finance does not establish reliable operation or wider local benefits (Steffen, 2018). The present study responds by examining, wherever records permit, the connection between financial arrangements, project implementation, electricity outcomes and the constraints that intervene between them. A third gap concerns the distinction between financial labels and environmental substance. Evidence from corporate green bonds suggests that credibility depends on disclosure, verification and the relationship between proceeds and actual environmental action (Flammer, 2021)

and the relationship between proceeds and actual environmental action (Flammer, 2021). In state level administrative practice, many relevant expenditures may not be labelled consistently, while some labelled activities may reveal little about environmental performance. cial records can therefore improve understanding of what should, and should not, be treated as green finance in Uttarakhand. The comparative treatment of sectors presents a fourth gap. Clean energy has attracted substantial research attention, but sustainable tourism, clean transportation and climate resilient agriculture have financing characteristics that Tourism research identifies environmental burdens across transport and supply chains, while agricultural research places greater emphasis on adaptation, local production conditions and food security. Bringing these sectors into a shared state level discussion can clarify where financial instruments, public support and evidence requirements diverge (Lenzen et al. , 2018). Another gap concerns the treatment of mountain risk in financial analysis. Research on Himalayan ecological change and the Chamoli disaster indicates that environmental vulnerability cannot be treated as an external background condition when infrastructure and livelihoods are exposed to physical hazards (Shugar et al. , 2021). Yet a project may appear financially attractive if its assessment excludes cumulative ecological effects, disaster exposure or the cost of maintaining services in difficult terrain. Integrating regional environmental conditions into the examination of financial suitability therefore strengthens the relevance of the analysis without turning the thesis into an independent disaster science study. The final gap is methodological and documentary. Existing state information is dispersed across policies, statistical publications, departmental records, regulatory documents and financial reports. Bowen (2009) argues that systematic document analysis requires attention to the origin, purpose, completeness and limitations of each record. In Uttarakhand, access to existing administrative material may be improved through applications under the Right to Information Act, but such material still needs verification and contextual interpretation. 1. 8 Rationale and Significance of the Study The rationale for the study follows from the economic importance of aligning financial decisions with environmental needs in a region characterised by ecological Green finance is frequently discussed as a national or international policy category, yet its practical value depends on how resources are Uttarakhand offers a setting in which the interaction among environmental risk, renewable energy, rural livelihoods, tourism and infrastructure can be examined without separating finance from its regional context. From an economic perspective, the study contributes to understanding how externalities, incomplete information, credit constraints and uncertain project returns influence

environmental investment. These concerns are especially relevant where the social benefits of a project extend beyond the borrower or where smaller applicants face difficulties in demonstrating repayment capacity. The analysis can help clarify why certain sectors attract finance more easily than others and why formal policy support does not necessarily produce equal access across districts or types of borrowers (Stiglitz & Weiss, 1981). The study also has significance for public policy. National climate commitments explicitly recognise the need for adaptation investment in vulnerable

A state level examination can indicate whether existing institutions, reporting systems and financial It may also identify where improved disclosure, coordination, project preparation or targeted support could assist the translation of public policy into documented action.

conditional on the quality of the evidence and do not imply that any proposed intervention has already been evaluated. Clean energy provides a particularly useful field

The sector contains identifiable technologies, formal policies, administrative agencies and official statistics on installed These features permit a structured examination of financial arrangements and reported outputs, even where complete project level financial information is unavailable.

renewable energy viability while recognising the environmental limits associated with project location, mountain hazards and electricity system requirements. A wider significance emerges from consideration of sustainable tourism, clean transportation and climate resilient agriculture. These sectors connect environmental investment with

Their financing requirements may differ from those of major electricity projects, particularly where enterprises are small, seasonal or geographically dispersed. the same state setting can reveal whether existing institutional arrangements appear sufficiently adaptable to varied investment needs. The study also has regional and distributional relevance. Official state records make it possible to examine differences among districts and between hill and plain areas, subject to the availability of comparable data (Directorate of Economics and Statistics, Government of Uttarakhand, 2025b). Such an approach helps prevent statewide averages from concealing unequal access to financial support or uneven exposure to environmental risk. It also directs attention to smaller producers, rural households and local enterprises whose investment constraints might be overlooked in assessments centred on large projects. Its final contribution concerns evidence and accountability. Distinguishing sanctioned amounts from disbursement,

installed capacity from generation, and policy announcements from implemented outcomes improves the quality of public discussion. Where existing records can be obtained under the Right to Information Act, the analysis may also clarify what information public institutions maintain and where important gaps remain (Government of India, 2005). The purpose is not to present administrative disclosure as an achievement in itself. It is to ground academic judgement in material that can be examined and, where necessary, qualified. 1. 9 Objectives of the Study The study is guided by the following objectives. To identify the potential of green finance in the study area. To identify the impact of green finance in clean energy sector with reference to Uttarakhand. To identify potential areas for growth in Green Finance in Uttarakhand. The first objective examines policies, institutions, financial instruments, sectoral activity and regional patterns to assess the documented basis for green investment in Uttarakhand. The second objective examines clean energy finance through available evidence on financial channels, project implementation, renewable electricity output and associated constraints. The third objective examines growth potential in Green Finance across renewable energy, sustainable tourism, clean transportation and climate resilient agriculture, using documented environmental relevance, financing requirements and institutional conditions. Table 1. 2 Alignment between objectives and documentary evidence Objective Principal analytical concern Indicative existing documentary evidence Objective 1 Potential and maturity of green finance in Uttarakhand. State policies, institutional mandates, schemes, budgets, available lending and district records. Objective 2 Role and reported effects of finance in the clean energy sector. Project sanctions and disbursements where available, installed capacity, generation and operational records. Objective 3 Potential areas for growth across the approved sectors. Sectoral policies, investment requirements, administrative records, financing constraints and reported initiatives. 1. 10 Hypotheses The hypotheses guide the examination of the maturity of Uttarakhand’s green-finance framework, the viability of clean energy as an area for financing, and the scope for growth in the identified sectors. Consistent with the exploratory and qualitative design, they will be assessed through descriptive comparisons

The assessment will distinguish evidence that supports a proposition from evidence that is contradictory, incomplete or unavailable.

examination. Hypothesis 1 H0: The development and implementation of green finance framework in Uttarakhand remains at a nascent stage, with significant variations across sectors and regions. H1: The development and implementation of green finance framework in Uttarakhand is at an evolved stage across sectors and regions. Hypothesis 2

H0: Clean energy does not present a viable sector for green finance in Uttarakhand. H1: Clean energy presents a viable sector for green finance in Uttarakhand. Hypothesis 3 H0: Uttarakhand does not hold substantial green finance growth potential in sectors like renewable energy, sustainable tourism, clean transportation and climate-resilient agriculture. H1: Uttarakhand holds substantial green finance growth potential in sectors like renewable energy, sustainable tourism, clean transportation and climate-resilient agriculture. The first hypothesis pair concerns institutional maturity and variation across sectors and regions. Relevant evidence may include policies, responsible institutions, available financial products, reporting practices, scheme coverage and district level distribution where records are accessible. The second pair concerns the viability of clean Its evaluation depends on evidence relating to financing arrangements, project development, electricity output, operating conditions and constraints. cy announcements exist. The third pair concerns the breadth of potential growth across the sectors named in the approved synopsis. Consideration of renewable energy, sustainable tourism, clean transportation and climate resilient agriculture remains conditional on documented opportunities, practical financial requirements and the limitations visible in available records. 1. 11 Scope and Delimitations of the Study The geographical scope of the study places India at the level of national policy and institutional context while identifying Uttarakhand as the principal study area. National legislation, climate commitments, financial regulation and development finance arrangements are examined because they influence the opportunities available within the state. The research does not attempt a comparative assessment of every Indian state, nor does it treat national financial totals as a substitute for Uttarakhand specific evidence. The substantive scope covers the potential of green finance, its documented role in the clean energy sector and potential areas for future growth in Uttarakhand. Clean and renewable energy receives particular attention as the second objective concerns that sector directly. Sustainable tourism, clean transportation and climate resilient agriculture are considered in relation to the third objective and the associated hypothesis. Supporting activities, such as energy storage, electricity networks, water conservation, waste management and charging infrastructure, are included where their relationship to the sectors is clear. The research follows an exploratory and qualitative design. Its principal materials consist of books, academic articles, official reports, policy documents, regulatory notifications, financial Where necessary, existing administrative documents was requested from public authorities under the Right to Information Act, 2005 (Government of India, 2005).

between stated targets, approved projects, sanctions, disbursements, installed capacity, generation and reported environmental or livelihood outcomes whenever the available In this study, relevant themes may include financial instruments, institutional responsibilities, sectoral allocation, regional variation, implementation barriers and reported outcomes. he description of trends and scale but does not transform the design into an econometric study. The term impact is used in the bounded sense appropriate to documentary evidence. It refers to reported or observable outputs and outcomes associated with financed activity, such as commissioned projects, renewable electricity generation, improved energy access or documented environmental benefits. Association should not Other influences, including public policy, technology prices, market conditions and local implementation capacity, may also contribute. s can support. Several delimitations follow from these choices. The study does not undertake original household surveys, structured interviews, experimental evaluation or It does not attempt to calculate an exhaustive total for all environmental investment if records are incomplete or Nor does it assume that every renewable project is environmentally suitable or that all rural development expenditure qualifies as green finance. nalysis remains sensitive to missing district data, inconsistent definitions, reporting delays, possible double counting and differences between project approval and operational status. 1. 12 Key Operational Definitions The following operational definitions establish how central terms are used in the study. They are intended to support consistent classification of documentary evidence rather than replace the specific definitions adopted by particular regulators or programmes. Green finance. Financial resources, instruments and institutional arrangements directed toward identifiable activities that reduce environmental harm, support climate mitigation or adaptation, protect ecological systems, or improve environmental resource use. Classification depends on the documented purpose and use of funds rather than on promotional language alone (Lindenberg, 2014). Climate finance. The component of environmental finance concerned specifically with reducing greenhouse gas emissions or strengthening resilience and adaptation to climate impacts. Climate finance is narrower than green finance because some environmental activities do not relate directly to climate mitigation or adaptation (United Nations Environment Programme, 2024). Sustainable finance. Financial activity that considers environmental, social and governance concerns together. A sustainable financial product is not automatically counted as green finance unless its environmental

purpose or environmentally relevant allocation can be established. Green investment. Expenditure on assets, projects, technologies or productive activities that generate identifiable environmental benefits. Green investment describes what is financed, while green finance describes the resources and financial arrangements that support the expenditure (Eyraud et al. , 2013). Clean energy. Energy systems and associated infrastructure that reduce environmental harm relative to conventional alternatives, including renewable electricity, decentralised solar applications, suitable small hydropower and energy efficiency measures, subject to assessment of their local ecological and social effects. Green finance potential. The documented or reasonably identifiable capacity for environmentally relevant financial support to be mobilised and applied within Uttarakhand, taking account of sectoral need, project opportunities, institutional arrangements, financial instruments, practical constraints and the availability of evidence. Financial flow. A recorded allocation, sanction, disbursement, investment or expenditure associated with an eligible activity. These categories are not interchangeable, and the stage represented by each figure must be identified before comparison. Impact of green finance. A documented output or outcome associated with environmentally relevant financing, including project commissioning, electricity generation, energy access, resource conservation or reported livelihood benefits. The term does not, by itself, imply statistically established causation. Project viability. The capacity of an investment to proceed and operate under identifiable technical, financial, institutional and environmental conditions. Viability is assessed through available evidence on costs, financing, expected or reported benefits, operational arrangements and relevant risks. Sustainable tourism. Tourism activity and related investment that seek to reduce environmental pressure, use energy and water more responsibly, manage waste appropriately and support local economic participation without treating visitor growth alone as evidence of sustainability (Lenzen et al. , 2018). Clean transportation. Transport systems, vehicles and supporting infrastructure that can reduce local pollution, energy intensity or greenhouse gas emissions, including suitable electric mobility and charging facilities, subject to operating conditions and the electricity supply available. Climate resilient agriculture. Agricultural practices, infrastructure and investment that strengthen the ability of farming systems to withstand climatic stress, protect production and improve the management of water, soil and related resources under local conditions (Lipper et al. , 2014). Study area. The state of Uttarakhand, considered through its relevant districts, sectors, institutions and environmental conditions. India provides the surrounding policy and financial context but does not replace Uttarakhand as the main unit of analysis. Existing administrative records. Documents, registers, reports, orders, financial statements or project records already held by public authorities. Where such

financial statements or project records already held by public authorities. Where such material is accessed through the Right to Information Act, it remains documentary evidence within the approved secondary source research design (Government of India, 2005). 1. 13 Organisation of the Thesis The thesis is organised into the five chapters. Their sequence moves from the statement of the problem and research design to the regional context, interpretation of evidence and the final synthesis of findings. Chapter 1, Introduction, establishes the background of the study, defines green finance and related concepts, explains the economic rationale for environmentally oriented investment, introduces the Indian and Uttarakhand contexts, and states the research problem, gap, significance, approved objectives and hypotheses. It also specifies the scope, delimitations, operational definitions and organisation of the thesis. Chapter 2, Review of Literature and Research Methodology, examines foundational and recent scholarship on green finance, environmental economics, renewable energy investment, sustainable tourism, clean transportation, climate resilient agriculture and Himalayan development. It then explains the exploratory qualitative research design, the use of secondary documentary sources, the treatment of existing administrative records obtained through the Right to Information Act, and the procedures for descriptive and thematic interpretation. Chapter 3, Brief Profile of Study Area, presents the geographical,

It describes the state's district structure, productive sectors, environmental pressures,

This chapter supplies the regional context needed to interpret differences in financial access, sectoral opportunity and project suitability.

l evidence and analysis chapter. It examines the existing framework and potential of green finance in Uttarakhand, including institutional arrangements, financial channels, sectoral and regional patterns. A substantive section within this chapter evaluates green finance in the clean energy sector by considering renewable and decentralised energy, electricity system requirements, policy support, financing channels, available project evidence, reported outcomes and constraints. The same chapter also considers potential growth in renewable energy, sustainable tourism, clean transportation and climate resilient agriculture. Chapter 5, Summary and Conclusion, brings together the principal findings in relation to the three approved objectives and the three hypothesis pairs. It discusses the economic and policy implications of the evidence, identifies the limitations of the study and indicates areas that may merit further research. A bibliography and relevant

This introductory chapter establishes that the value of green finance in Uttarakhand cannot be inferred from environmental ambition or financial terminology alone.

assessed through the relationship among institutional arrangements, actual financial support, sectoral conditions, regional access and documented results. The next chapter develops the scholarly and methodological foundation required for that assessment.

Reference List Akerlof, G. A. (1970). The market for “lemons”: Quality uncertainty and the market mechanism. The Quarterly Journal of Economics, 84(3), 488–500. https://doi. org/10. 2307/1879431 Banga, J. (2019). The green bond market: A potential source of climate finance for developing countries. Journal of Sustainable Finance & Investment, 9(1), 17–32. https://doi. org/10. 1080/20430795. 2018. 1498617 Coase, R. H. (1960). The problem of social cost. The Journal of Law and Economics, 3, 1–44. https://doi. org/10. 1086/466560 Department of Economic Affairs, Government of India. (2022). Framework for sovereign green bonds. https://dea. gov. in/framework-sovereign-green-bonds Directorate of Economics and Statistics, Government of Uttarakhand. (2025a). Economic survey 2024–25. https://des. uk. gov. in/economic-survey/ Directorate of Economics and Statistics, Government of Uttarakhand. (2025b). Uttarakhand at a glance 2024–25. https://des. uk. gov. in/document-category/uttarakhand-at-a-glance/ Egli, F. , Steffen, B. , & Schmidt, T. S. (2018). A dynamic analysis of financing conditions for renewable energy technologies. Nature Energy, 3(12), 1084–1092. https://doi. org/10. 1038/s41560-018-0277-y Eyraud, L. , Clements, B. , & Wane, A. (2013). Green investment: Trends and determinants. Energy Policy, 60, 852–865. https://doi. org/10. 1016/j. enpol. 2013. 04. 039 Flammer, C. (2021). Corporate green bonds. Journal of Financial Economics, 142(2), 499–516. https://doi. org/10. 1016/j. jfineco. 2021. 01. 010 Geddes, A. , Schmidt, T. S. , & Steffen, B. (2018). The multiple roles of state investment banks in low-carbon energy finance: An analysis of Australia, the UK and Germany. Energy Policy, 115, 158–170. https://doi. org/10. 1016/j. enpol. 2018. 01. 009 Gillingham, K. , & Stock, J. H. (2018). The cost of reducing greenhouse gas emissions. Journal of Economic Perspectives, 32(4), 53–72. https://doi. org/10. 1257/jep. 32. 4. 53 Government of India. (2005). The Right to Information Act, 2005. https://www. indiacode. nic. in/handle/123456789/2065 Government of India. (2022). India’s updated first nationally determined contribution under the Paris Agreement. https://unfccc. int/sites/default/files/NDC/2022-08/India%20Updated%20First%20Nationally%20Determined%20Contrib. pdf Government of Uttarakhand. (2019). Uttarakhand electric vehicle policy 2019. https://startuputtarakhand. uk. gov. in/single_state_notifications/47/Uttarakhand%20Electric%20Vehicle%20Policy%202019 Immerzeel, W. W. , Lutz, A. F. , Andrade, M. , Bahl, A. , Biemans, H. , Bolch, T. , Hyde, S. , Brumby, S. , Davies, B. J. , Elmore, A. C. , Emmer, A. , Feng, M. , Fernández, A. ,

Haritashya, U. , Kargel, J. S. , Koppes, M. , Kraaijenbrink, P. D. A. , Kulkarni, A. V. , Mayewski, P. A. , … Baillie, J. E. M. (2020). Importance and vulnerability of the world's water towers. Nature, 577(7790), 364–369. https://doi. org/10. 1038/s41586-019-1822-y Intergovernmental Panel on Climate Change. (2023). Climate change 2023: Synthesis report. IPCC. https://www. ipcc. ch/report/ar6/syr/ International Energy Agency. (2024). Global EV outlook 2024. https://www. iea. org/reports/global-ev-outlook-2024 Lenzen, M. , Sun, Y. Y. , Faturay, F. , Ting, Y. P. , Geschke, A. , & Malik, A. (2018). The carbon footprint of global tourism. Nature Climate Change, 8(6), 522–528. https://doi. org/10. 1038/s41558-018-0141-x Lindenberg, N. (2014). Definition of green finance. German Development Institute. https://www. idos-research. de/en/publications/mitarbeiter-sonstige/article/definition-of-green-finance/ Lipper, L. , Thornton, P. , Campbell, B. M. , Baedeker, T. , Braimoh, A. , Bwalya, M. , Caron, P. , Cattaneo, A. , Garrity, D. , Henry, K. , Hottle, R. , Jackson, L. , Jarvis, A. , Kossam, F. , Mann, W. , McCarthy, N. , Meybeck, A. , Neufeldt, H. , Remington, T. , … Torquebiau, E. F. (2014). Climate-smart agriculture for food security. Nature Climate Change, 4(12), 1068–1072. https://doi. org/10. 1038/nclimate2437 Mazzucato, M. , & Semieniuk, G. (2018). Financing renewable energy: Who is financing what and why it matters. Technological Forecasting and Social Change, 127, 8–22. https://doi. org/10. 1016/j. techfore. 2017. 05. 021 National Bank for Agriculture and Rural Development. (2025). Annual report 2024–25. https://www. nabard. org/financialreport. aspx?cid=505&id=24 Ozili, P. K. (2022). Green finance research around the world: A review of literature. International Journal of Green Economics, 16(1), 56–75. https://doi. org/10. 1504/IJGE. 2022. 125554 Polzin, F. , Egli, F. , Steffen, B. , & Schmidt, T. S. (2019). How do policies mobilize private finance for renewable energy? A systematic review with an investor perspective. Applied Energy, 236, 1249–1268. https://doi. org/10. 1016/j. apenergy. 2018. 11. 098 Reserve Bank of India. (2023a). Framework for acceptance of green deposits. https://www. rbi. org. in/Scripts/NotificationUser. aspx?Id=12487&Mode=0 Reserve Bank of India. (2023b). Report on currency and finance 2022–23: Towards a greener cleaner India. https://rbidocs. rbi. org. in/rdocs/Publications/PDFs/RCF03052023395FAF37181E40188BAD3AFA59BF3907. PDF Securities and Exchange Board of India. (2023). Revised disclosure requirements for issuance and listing of green debt securities. https://www. sebi. gov. in/legal/circulars/feb-2023/revised-disclosure-requirements-for-issuance-and-listing-of-green-debt-securities_67837. html Securities and Exchange Board of India. (2025). Framework for environment, social and governance (ESG) debt securities (other than green debt securities). https://www. sebi. gov. in/legal/circulars/jun-2025/framework-

for-environment-social-and-governance-esg-debt-securities-other-than-green-debt-securities-_94424. html Shugar, D. H. , Jacquemart, M. , Shean, D. , Bhushan, S. , Upadhyay, K. , Sattar, A. , Schwanghart, W. , McBride, S. , de Vries, M. V. W. , Mergili, M. , Emmer, A. , Deschamps-Berger, C. , McDonnell, M. , Bhambri, R. , Allen, S. , Berthier, E. , Carrivick, J. L. , Clague, J. J. , Dokukin, M. , … Westoby, M. J. (2021). A massive rock and ice avalanche caused the 2021 disaster at Chamoli, Indian Himalaya. Science, 373(6552), 300–306. https://doi. org/10. 1126/science. abh4455 Steffen, B. (2018). The importance of project finance for renewable energy projects. Energy Economics, 69, 280–294. https://doi. org/10. 1016/j. eneco. 2017. 11. 006 Stern, N. (2007). The economics of climate change: The Stern review. Cambridge University Press. https://www. cambridge. org/core/books/economics-of-climate-change/A1E0BBF2F0ED8E2E4142A9C878052204 Stiglitz, J. E. , & Weiss, A. (1981). Credit rationing in markets with imperfect information. The American Economic Review, 71(3), 393–410. https://www. jstor. org/stable/1802787 United Nations. (2015). Transforming our world: The 2030 agenda for sustainable development. https://sdgs. un. org/2030agenda United Nations Environment Programme. (2024). Adaptation gap report 2024: Come hell and high water. https://www. unep. org/resources/adaptation-gap-report-2024 Uttarakhand Renewable Energy Development Agency. (n. d.). Hydro energy schemes. https://ureda. uk. gov. in/hydro-energy-schemes/ Uttarakhand Renewable Energy Development Agency. (2015). Policy for development of micro and mini hydro power projects up to 2 MW. https://ureda. uk. gov. in/hydro-energy-schemes/ Uttarakhand Renewable Energy Development Agency. (2023). Uttarakhand state solar policy 2023. https://ureda. uk. gov. in/solar-energy-schemes/ Wang, Y. , & Zhi, Q. (2016). The role of green finance in environmental protection: Two aspects of market mechanism and policies. Energy Procedia, 104, 311–316. https://doi. org/10. 1016/j. egypro. 2016. 12. 053 Wester, P. , Mishra, A. , Mukherji, A. , & Shrestha, A. B. (Eds.). (2019). The Hindu Kush Himalaya assessment: Mountains, climate change, sustainability and people. Springer. https://doi. org/10. 1007/978-3-319-92288-1 Zerbib, O. D. (2019). The effect of pro-environmental preferences on bond prices: Evidence from green bonds. Journal of Banking & Finance, 98, 39–60. https://doi. org/10. 1016/j. jbankfin. 2018. 10. 012